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Tackling the Complexity of Emergency Response Systems: Creating Transport-Focused Digital Twins

Fabian Schuhmann *, Moritz Sturm, Till Zacher and Markus Lienkamp

Institute of Automotive Technology, Department of Mobility Systems Engineering, TUM School of Engineering and Design, Technical University of Munich, 85748 Garching, Germany
* Correspondence: fabian.schuhmann@tum.de

Highlights

What are the main findings?

- The proposed framework builds transport-focused digital twins of emergency response systems, combining multiple data sources. The combination of real-time traffic data, GPS data from emergency vehicles (EVs), and road network information enables precise network analysis, e.g., identifying bottlenecks where EVs are affected by traffic. Our data shows that more favorable traffic conditions and more spacious road structures are positively associated with improved EV progress in urban environments.
- The agent-based, mesoscopic model of the Munich Fire Department shows good results in validation compared to historical incident data.

What is the implication of the main finding?

- Digital twins can be used to assess the impact of the transport system on the emergency response system and to systematically simulate future 'what-if' scenarios.
- Our data suggests that the transition to sustainable mobility can have a positive impact on emergency services. However, additional datasets and research are needed to develop specific recommendations for action.

Abstract

Providing medical and technical assistance to people in life-threatening situations requires the coordinated cooperation of numerous actors within the emergency response system. The efficiency of the emergency response system is thereby influenced by the transport infrastructure and the traffic conditions. Organizations and authorities with safety responsibilities are increasingly faced with the challenge of assessing the impact of changes to the transport system on the overall system's effectiveness. The overall objective of this paper is to develop an efficient and cost-effective simulation and analysis platform for generating transport-focused digital twins, enabling organizations and authorities to monitor the current emergency response system and digitally analyze various 'what-if' scenarios for future planning. Our model combines various data sources, including real-time traffic data, recorded GPS data from emergency vehicles (EVs), and the road network. The data serves as the foundation for the indicator-based network analysis and the system model. The main actors in the emergency response system are modeled in the agent-based model to analyze the spatiotemporal impact of changes in the transport system on the system's effectiveness. The developed simulation and analysis platform is applied to a case study of the Munich Fire Department, Germany. First, a network analysis using regression of EV speed on reported real-time traffic speed helps identify problematic areas where EVs are affected by traffic. Secondly, the agent-based model of the Munich fire department demonstrates good validation results against historical incident data, with recorded trajectory data used

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for model calibration. Our work contributes to efficient, data-driven planning for future emergency response systems.

Keywords: emergency response system; emergency service; digital twin; GPS trajectory analysis; agent-based emergency model

1. Introduction

The transition to sustainable mobility is changing the current status quo of our transport system and challenging existing norms and circumstances [1]. The mobility transition introduces various changes, such as the implementation of $30\frac{\text{km}}{\text{h}}$ zones and modal filters, redesigned intersections [2], and the reduction of the number of traffic lanes. These changes affect emergency vehicles (EVs) as they respond to incidents in various ways, e.g., through reduced driving speed , additional delays at intersections , altered or restricted turning maneuvers, and potential bottlenecks on single-lane roads without passing opportunities. Nevertheless, it must be ensured that help reaches patients immediately.

From the perspective of emergency service operators, three key questions arise in the context of the mobility transition: (i) What impact do traffic conditions and road cross-section characteristics have on the driving speed of EVs? (ii) Which edges in the road network require expert assessment as critical situations arise frequently? (iii) How do changes in average speeds and accessibility influence the emergency response system on a system level?

A primary challenge in addressing these questions is the lack of practical, data-driven tools for monitoring and planning, which currently limits the empirical impact assessments of transport system changes on the emergency response system. Recent smart-city research is increasingly using data-driven models to enhance urban emergency operations. For instance, recent work optimizes emergency distribution systems explicitly targeting spatial equity in communities with differing mobility characteristics [3], while another work demonstrates algorithmic allocation of emergency resources in a fire-following-earthquake scenario [4]. Additionally, Sadler [5] highlights the importance of geographically fine-grained planning. Together, these perspectives underscore the need for transport-focused digital twins capable of quantifying how network-level design and operational changes influence the emergency response system.

Our work aims to develop an efficient and cost-effective simulation and analysis platform for creating transport-focused digital twins of emergency response systems. This toolchain is designed to support authorities and organizations with safety responsibilities in analyzing the current system and in developing data-driven plans for future configurations. Our solution is based on linking open data with locally collected information, such as GPS data. Thereby, EVs are used as mobile sensors to collect the necessary model parameters. Our framework addresses the three key questions mentioned above by (i) performing a regression analysis between real-time floating car data and EV GPS data, (ii) defining a combined criticality indicator, and (iii) developing an agent-based model for system simulation. The contribution of this work to the state of the art lies in its practical and data-driven support for transport-related monitoring and planning of emergency response systems.

The developed framework is applied in a case study with the Munich Fire Department. Recorded trajectory data is analyzed to identify problem areas in the road network using regression analyses and the defined criticality indicator. Additionally, a dynamic system model is calibrated and validated using historical incident data.

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The remaining parts of this paper are structured as follows: Section 2 outlines the current state of the art and highlights the research gap. Section 3 describes the methodology behind the analysis framework, the data acquisition, the developed indicators, and the agent-based model. Section 4 presents the case study, the results of which are presented in Section 5. Section 6 discusses the developed framework and its findings.

2. State of the Art

Various models have been developed in the literature to investigate the effects of changes and extreme situations on the emergency response system, such as the model presented in [6]. These models differ in terms of the level of detail they use in modeling the transport. The following chapter first provides an overview of recent research work that uses detailed open-source transport models, followed by an examination of studies that focus on GPS-based data analysis for EVs.

2.1. Emergency Service Specific Extensions in Transport Simulation Models

Bieker-Walz et al. [7] implemented special rights for EVs in SUMO [8] (e.g., passing red lights, cars forming an emergency lane) and compared simulated EV driving behavior to real-world data. The focus of the work is on modeling an EV's interaction with traffic at a detailed level for a single intersection, finding that the simulation lets ambulances pass intersections slightly faster than in reality.

Soni and Weronek [9] modeled an EV with surrounding traffic on a busy street in Frankfurt, Germany, with various traffic signal preemption strategies as well as lane closure scenarios using SUMO [8]. They used the EV travel times and delays as key metrics. By comparing the baseline versus several scenarios, they found that giving the EV signal priority on their test corridor significantly reduced total trip time.

Capodieci et al. [10] extended MATSim [11] to simulate a smart city scenario in which cars are equipped with Advanced Driver Assistance Systems (ADAS) and have the ability to communicate via vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) connections. They modeled traffic accidents and tested various mitigation strategies. In their simulations, connected and semi-autonomous vehicles were guided to yield or reroute, which improved emergency response times to incidents. Their findings indicate that when cars have different communication capabilities, EVs reach incident sites faster.

Koch et al. [12] simulated ambulance dispatch in New Windsor (NY) under disaster conditions, such as road network disruptions. Their agent-based model examined how the loss of key road links (e.g., due to floods) slows overall emergency response. A key finding from their data was that certain roads are critical: removing them (network disruptions) led to an increase in the city's overall response times.

Bandyopadhyay and Singh [13,14] developed an agent-based simulation for Allahabad city using GPS-based EV driving data to calibrate the road network. The multi-agent system is implemented in GAMA [15]. A multi-criteria weighted road network is used for routing EVs and the agent-based model comprises fire station, EVs, and incidents. The validation is performed by comparing simulated and actual response times. An experimental analysis is performed at the system level, focusing on an organizational change in [14].

Our previous work [16] presents a strategic model to enable dynamic infrastructure planning by emergency services based on the SUMO traffic simulation by modeling incidents and emergency service structures at the city scale. The simulation model allows infrastructure and capacity scenarios to be simulated in order to systematically compare their effects on accessibility and quality of service.

Wachter et al. [17] conduct a time comparison between simulated ground-based ambulances and calculated horizontal flight paths of eVTOLs in the Stavanger metropolitan

region and its surrounding islands using a calibrated, microscopic SUMO traffic model. They systematically evaluate ambulance travel times from the hospital to all possible locations within the simulation area, using a predefined grid.

Mínguez et al. [18] present SEMSIM, a simulation tool methodologically comparable to [16], which is based on SUMO and was designed to optimize the location selection of ambulances and their deployment configuration in the Terres de l'Ebre region of Catalonia. Building on a microscopic SUMO simulation, external tools are used to simulate ambulance journeys. As part of the validation process, four stations with four ambulances are modeled and validated against real data.

The state of the art for emergency service specific extensions in transport simulation models is summarized in Table 1.

Table 1. Literature classification (**Legend:** ○ = not available, ◔ = low, ◑ = medium, ◕ = high, ● = comprehensive; *Use Case Type*: R = regular operation, D = disaster; *Emergency Service*: A = Ambulance, FD = Fire Department, VFD = Voluntary Fire Department, † = conceptionally).

Source	Year	Region	Number of Stations	Emergency Service	Use Case Type	Level of Detail: Emergency Data	Agent-Based System Model	Level of Detail: Transport Model	Capability: System Monitoring	Applicability by Operators
[7]	2018	Brunswick, DE	n.a.	n.a.	R	●	○	●	○	○
[9]	2023	Frankfurt, DE	n.a.	n.a.	R	○	○	●	○	○
[10]	2021	Modena, IT	n.a.	A	R	○	◔	◔	○	○
[12]	2020	New Windsor, US	1	A	D	◔	◑	◔	○	○
[13,14]	2016/2018	Allahabad, IN	1	FD	D	◕	◕	◔	◑	◔
[17]	2024	Stavanger, NO	1	A	R	◑	◑	●	○	○
[16]	2024	Munich, DE	10	FD	R	◑	◑	◑	○	◑
[18]	2025	Catalonia, ES	4	A	R	◑	◕	◑	○	◔
This Work	2025	Munich, DE	10	FD, A, VFD †	R	●	●	◑	●	●

2.2. Commercially Available Software

In addition to scientific work, commercial software solutions also provide digital twins for emergency response systems, e.g., [19] or [20]. However, the underlying methodology is generally not publicly disclosed, which limits the application of commercial software in academic environments. To the best of our knowledge, agent-based models for simulating the holistic dynamic response of the emergency response system have not been reported.

2.3. GPS-Based Data Analysis for EVs

The speed at which EVs travel in urban and rural areas is a subject of ongoing research. The following provides an overview of existing studies and the parameters that have been examined to date. Bandyopadhyay and Singh [13] provide mean speeds and standard deviations of fire EVs for different classes of road segments within the Indian city Allahabad. The study is based on a total of 35 recorded GPS trajectories. Lupa et al. [21] analyzed GPS records from 300 ambulances in Lesser Poland to derive real ambulance speed characteristics and examine how factors such as time of day, traffic, built-up areas, and the use of sirens affect travel times. Pappinen and Nordquist [22] compared ambulance driving speeds in urgent and non-urgent situations in Finland, using GPS data from 2018, and analyzed them against normal and winter speed limits. Bieker-Walz et al. [23] examined GPS data from the Brunswick Fire Department, which was recorded by 24 EVs over a

period of five years. In particular, they analyzed the spatial and temporal distribution as well as the duration and length of the emergency trips. Additionally, various parameters that influence the route selection of EVs were examined. Steinvoord [24] analyzes a data set of GPS recordings from the German emergency medical service stations in Bad Oldesloe and Reinfeld. The data was collected over a period of two months at a frequency of 1 Hz and evaluated in terms of location, road category, and vehicle category. In [25] we first presented our study for data recording, now covering 18 months of data from the Munich Fire Department. In our previous work, we have examined system parameters, speeds by road category, and the route selection parameters of EVs.

2.4. How rescuePY Stands out

Despite the availability of several scientific publications and commercial systems, to the best of our knowledge, there are still gaps in research: (i) There is no open, expandable end-to-end simulation and analysis platform that supports monitoring and analyzing the interaction between the transport system and the emergency response system, as well as the prediction of system-wide impact. (ii) Existing simulation models focus either on emergency medical services or fire departments, i.e., one sub-area of the overall emergency response system. Volunteer organizations have not been sufficiently considered in simulation approaches, and a holistic, agent-based system model is currently lacking. (iii) Data sets describing the emergency trips made by EVs in urban (European) environments are rare. While the average speed and spatial distribution have already been studied, there has been no analysis of the dependence on surrounding traffic using floating-car data (FCD), nor has there been any indicator for identifying critical road segments in the network based on the records.

To overcome these limitations, rescuePY has been developed as a holistic, open-source-based data analysis and simulation platform that integrates heterogeneous data sources (e.g., GPS, IMU, real-time floating car data, and land-use information), combines indicator-based evaluation with agent-based simulation in a two-level analytical framework, provides a modular open-source architecture, and supports system-scale scenario analyses.

3. Methods

In previous work, relevant components for the simulation and analysis platform have already been developed and published. The idea for the simulation model and a first implementation is presented in [16,26]; the GPS data analysis is presented in [25,27], focused on route choice modeling and general route characteristics, including average speeds. This publication aims to establish the relationship between the various components and significantly expand the simulation model and data analysis.

Our framework integrates various data sources to assess the relationship between the transport system and the emergency response system and evaluate potential future scenarios. Figure 1 shows the overall architecture of the developed framework. The core of the data analysis framework is the automated collection and historization of data sources such as real-time traffic data [28,29], GPS measurements, land use and street network [30] and emergency response system data. This data is imported into a consolidated data model based on PostGIS [31] and forms the basis for further analysis. Using pgRouting [32] for static and dynamic network analyses and SUMO [8] for dynamic system models, the current system and future planning scenarios can be evaluated. Using SUMO [8] as the basis for the transport simulation has the advantage that proven tools for data processing can be used, the simulation can be easily visualised, and mesoscopic and microscopic traffic models can be integrated. The underlying software architecture is modular and scalable, enabling it to meet varying requirements and ensuring easy expansion. All parameters

can be adjusted manually via the user interface, allowing for simple adaptation to local requirements. Our model is (partially) open source and available [33].

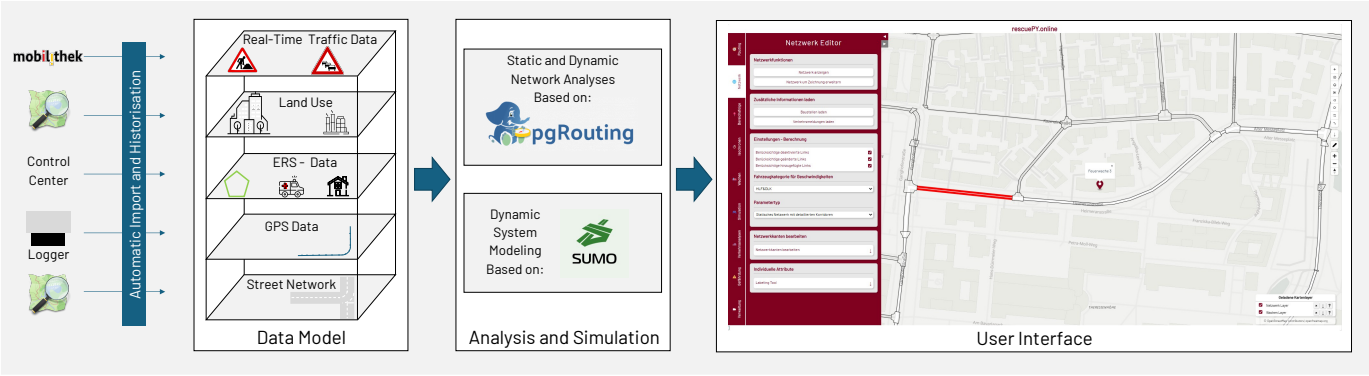


Figure 1. Architecture of the rescuePY platform (adapted from our own figure previously published in [34]; Map: created with OpenFreeMap ©OpenMapTiles and data from OpenStreetMap [30]; mobilithek: [28]; SUMO: [8]; pgRouting: [32]).

3.1. Data Analysis

This subsection consists of two parts. The first section describes the data analysis pipeline that enables the recorded data to be referenced with the network. The second section defines the indicators to be analysed.

3.1.1. Data Analysis Pipeline

The data analysis pipeline is described in detail in [27]. The pipeline is summarized below and is shown graphically in Figure 2:

- **Step 1—Calibration:** Transformation of IMU data into the vehicle coordinate system according to [35], supplemented by smoothing.
- **Step 2—Preparation:** Aggregation of GPS points to emergency trips (speed $> 1\frac{\mathrm{m}}{\mathrm{s}}$). Short interruptions of up to 180 s, e.g., due to tunnels or traffic jams, are permitted and will be taken into account.
- **Step 3—MapMatching:** Since EVs have special privileges (driving against the direction of traffic, exceeding speed limits, ignoring turn restrictions), the approach of [36] was adapted for our use case. The result is a sequence of edges that were traversed by the EV.
- **Step 4—Assignment:** Assignment of GPS points to edge sections based on geometric criteria (next edge $<$ 30 m, same direction, and ensuring that the projection lies within the edge). The chosen two-step process ensures quality and efficiency.
- **Step 5—Filter:** Filtering by signal quality (number of satellites, DOP), start at the station, end at the location of incident/operation, and comparison with the control center protocol logs.

The resulting trajectories consist of edge-referenced GPS measurement points. Based on the referenced GPS measurement points, average speeds per edge and time losses at junctions can be calculated.

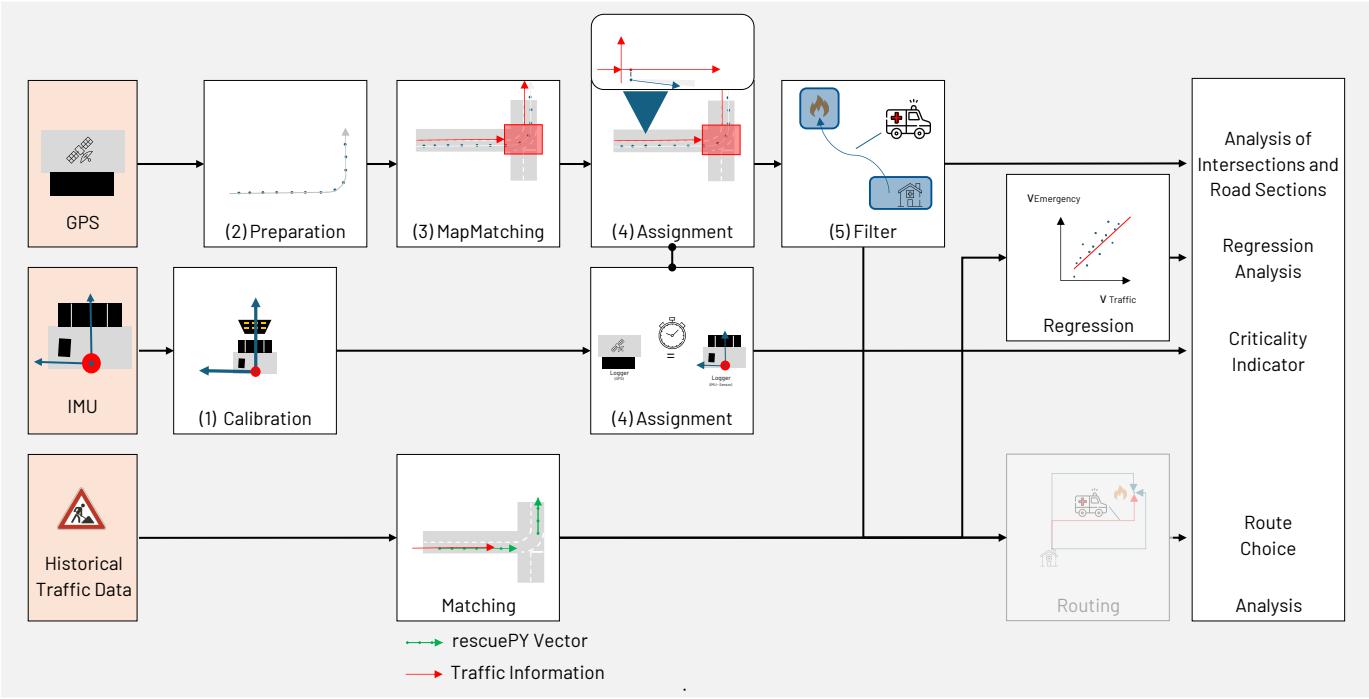


Figure 2. Data Analysis Pipeline of rescuePY.

3.1.2. Definition of Indicators

In urban environments, monitoring large road networks is complex, and identifying potentially critical road sections can be difficult. To support the road network monitoring by emergency services, we propose two indicators, which are presented below:

Criticality Indicator: To identify potentially critical edges within the network for EVs, multiple indicators are used to analyze the recorded data of the data logger. Individual road sections are evaluated based on the frequency of critical events x_{crit} relative to the total number of passings n, as shown in Equation (1):

$$R_{\text{Roadsegment}} = \frac{\sum x_{\text{crit}}}{\sum n}. \tag{1}$$

The indicator x_{crit} is a compound indicator that aggregates multiple individual indicators, which are adapted from literature into one comprehensive location-fixed indicator. As outlined in Equation (2), the indicator is set to 1 if any of the individual indicators are triggered; otherwise, it is set to 0.

$$x_{\text{crit}} = \begin{cases} 1, & \text{if } |v| \leq 2\frac{\text{m}}{\text{s}} \ \vee \ a_x \geq 3\frac{\text{m}}{\text{s}^2} \ \vee \ a_y \geq 3\frac{\text{m}}{\text{s}^2} \ \vee \ |\dot{a}_x| \geq 10\frac{\text{m}}{\text{s}^3} \ \vee \ |\dot{a}_y| \geq 10\frac{\text{m}}{\text{s}^3} \\ 0, & \text{else.} \end{cases} \tag{2}$$

The thresholds in Equation (2) are used as conservative limits. Several reference values originate from passenger-car studies. We transfer them as a lower bound for criticality. EVs are heavier, often have a higher center of gravity, and may carry sensitive cargo, which typically reduces the acceptable acceleration and jerk range. Thus, maneuvers that reach passenger-car "aggressive" or "near-crash" levels are likely to be critical. This conservative setup reduces false positives when flagging segments, but it may under-detect less extreme maneuvers that could still matter for emergency operations. Therefore, our model is fully adaptable, allowing thresholds to be set on an individual level.

The first indicator captures *low speed events* where the vehicle's speed drops below a critical threshold v_{crit}. We selected a threshold of $2\,\frac{\text{m}}{\text{s}}$, as going below this speed is

generally not expected in normal traffic conditions and thus signals a hindrance to the EV's progress. The second and third indicators capture *acceleration peaks* in the longitudinal and lateral directions, respectively. As described by the US NHTSA, a near-crash situation is characterized by a rapid evasive maneuver, indicated by an acceleration or deceleration that approaches the limits of the vehicle's capabilities [37]. They consider longitudinal accelerations of $a_x \geq 4.9\,\frac{m}{s^2}$ and lateral accelerations of $\geq 3.9\,\frac{m}{s^2}$ as near-crash situations [37]. Johnson and Trivedi identified that the average peak lateral acceleration during non-aggressive left and right turns is approximately 0.30 g and therefore remains below $a_y < 3\,\frac{m}{s^2}$ [38]. Serre et al. measured that accelerations inside an ambulance during emergency driving are below $a = 3\,\frac{m}{s^2}$ for 98.6 % of the time and investigated anything above that as a potential hazard [39]. Based on these findings, we set the thresholds for our indicators to $a_x \geq 3\,\frac{m}{s^2}$ and $a_y \geq 3\,\frac{m}{s^2}$. As Bagdadi identified, high jerk values as a strong indicator for near-crash situations [40], we also include a *jerk* value as part of our compound indicator. Jerk measures the rate of change in acceleration [40]. Previous work has used jerk thresholds of $10\,\frac{m}{s^3}$ [40] to identify critical situations. In line with previous research we choose jerk values of $|\dot{a}_x| \geq 10\,\frac{m}{s^3}$ and $|\dot{a}_y| \geq 10\,\frac{m}{s^3}$ as thresholds for criticality.

Regression Analysis: In addition to the kinematics-based criticality indicators, we also suggest a *traffic speed regression* indicator. As shown in previous research, the interaction between EVs and other traffic is a key factor in determining the overall response time of an EV [21,41,42]. Interactions may include the evasive behavior of other vehicles and the effects of traffic jams on the speed of the EV. To capture these effects, we propose a regression analysis (see Figure 3), in which we correlate the EV's speed for a specific section of road with the nearest available reported average speed of all other vehicles over several journeys on the same section of road. A positive association between the EV's speed and the surrounding traffic speed would be consistent with the EV being constrained by ambient traffic conditions. We therefore treat such segments as candidates for manual review. We estimate a linear regression on each road segment using a Theil–Sen regressor. This estimator is chosen for its robustness against outliers. To quantify the correlation, we use the Pearson correlation coefficient r. We define street segments as relevant for manual review, having a Pearson correlation coefficient $r > 0.2$ that is statistically significant at the 5% level and a slope $m > 0.5$.

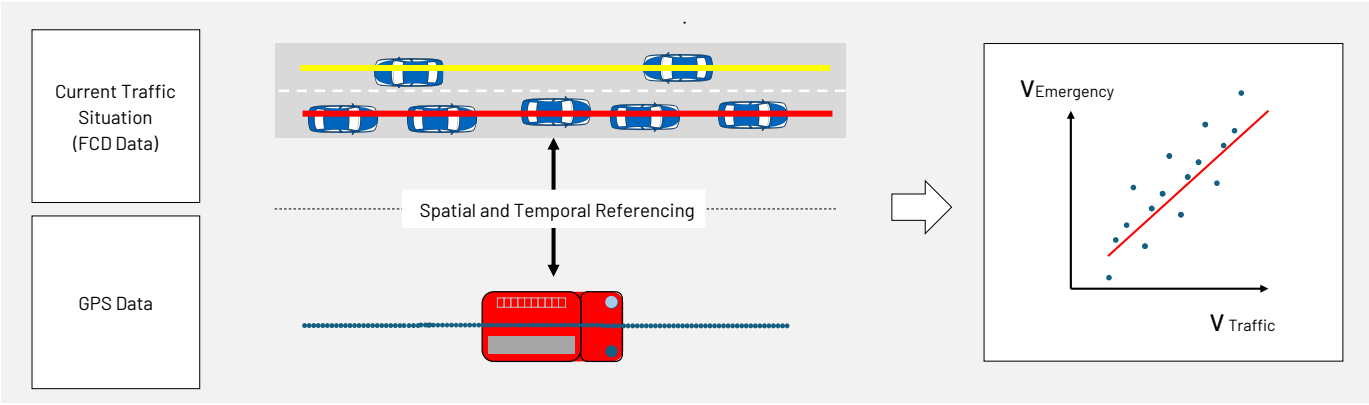


Figure 3. Referencing the current traffic situation and GPS-based EV speed.

3.2. Dynamic System Model

Profound changes in accessibility within the city necessitate a comprehensive assessment that considers the entire emergency response system and its dynamic response to incidents. To enable this, our simulation and analysis platform includes a dynamic system model. The model follows an agent-based approach: each actor of the emergency response

system is modeled as an independent agent with specific attributes and state transitions. The system dynamics result from the interaction of these agents.

3.2.1. Modeled Agents, Additional Components and Their Properties

The simulation replicates the real process by assigning specific tasks to each type of emergency service: career firefighters, volunteer firefighters and emergency medical services. All emergency services follow a similar basic sequence: gearing up, traveling along a realistic route, responding on site, and then returning to their station. Volunteer firefighters first drive from their homes to the station, while ambulances may drive to the nearest hospital before returning to the station. Once an EV has left the scene, it is available for the next incident.

In expert discussions, the entities to be modeled and their essential characteristics were defined. The relevant services include career fire departments, volunteer fire departments, and emergency medical services.

Modeled Agents:

- **Hospital**: A hospital treats patients arriving by ambulance. The hospitals are characterized by a defined capacity and a fixed location.
- **Fire station**: Fire stations are responsible for managing the allocation of crew members to specific roles and vehicles. They also serve as the home base for all EVs. Each fire station has a defined location, and every EV is assigned to exactly one station, to which it returns after each operation. Stations are assigned specific emergency capabilities.
- **Dispatch Agent**: The dispatch agent becomes active when an incident occurs. The dispatch logic corresponds to the real system. Based on predefined zones and the pre-calculated order of fire stations, it is determined which station should respond, given the current status of the EVs and the required resources. The incident is then assigned accordingly.
- **Emergency Vehicles**: EVs are vehicle-agents. Every EV has a home base, to which it returns after completing an operation. They each have a type, which indicates different capabilities. This is needed to assign the required EV for each emergency type. Ambulances are specifically designed to receive and transport a single patient. Each EV has a list of currently assigned personnel and a status reflecting its current position/task. The most important statuses can be described as follows: In status 2, the EV is at its base. Once assigned to an emergency and ready for deployment, it enters status 3 and proceeds to the incident location. Upon arrival and engagement, it is in status 4. After completing the operation, it switches to status 1 and returns to its home base.

Additional Components:

- **Incident**: An incident is an emergency event defined prior to the simulation run, which is triggered at a predefined time. Each incident has a location and a classification.
- **Person**: A person can either be a member of an EVs crew or a patient. Crew members possess a set of roles they are qualified to perform, as well as an active role they currently fulfill. If a crew member belongs to a volunteer fire station, an additional parameter is defined: a radius indicating the distance from which the member can be randomly dispatched to the station in case of an emergency.
- **Log Entry**: Every status change of an EV, as well as every trip, generates a log entry that can be used for a more detailed analysis of the simulation results.
- **Zones**: The simulated area of interest is divided into multiple smaller zones, which are required for the station order. These zones must be generated or provided as input before executing the simulation.

- **Station Order**: The station order defines, based on the zone in which an emergency occurs, the order of the nearby stations to respond to an emergency in that zone. The first available station with the required equipment from the calculated order is selected. This configuration must be defined prior to running the simulation.
- **Alarm and Dispatch Order**: The alarm and dispatch order specifies the required number and type of equipment and personnel for each type of incident. This configuration must be defined before the simulation begins.

3.2.2. Implementation of the Transport Model

The basic modeling and interaction with the SUMO simulation [8] is taken from [16]. A mesoscopic SUMO simulation is used to simulate emergency response journeys. Vehicles in motion (on their way to or from the scene) and at the scene are represented in the SUMO simulation. Vehicles on standby at the station are not included in the simulation. The interaction logic with the traffic simulation is implemented in TraCi. The driving speeds of EVs are determined based on average speed values from the GPS data. To enable a city-wide simulation with reasonable compute time, surrounding vehicles are not explicitly simulated, but their effects are taken into account via reduced average speeds. This reduction can be modeled on a section-by-section basis or by road category. The present simulation model covers public traffic areas accessible to passenger cars. It focuses on the movement of EVs in the road space, starting with their departure from the station and ending with their arrival at the scene of the emergency, defined as the parking of the vehicle in the public road space. Missing turn relationships, which exist due to the connection of edges and nodes but for which no connection exists, are supplemented for EVs using netconvert. The routing method used for EVs is the logic described in [27], but for reasons of efficiency, it is limited to the fastest route only.

3.2.3. Modeled Processes

As soon as an incident occurs during the simulation, the relevant EVs are immediately dispatched. The selection of EVs depends on the incident type, the alarm and dispatch order, and the location of the incident. The dispatching can be configured to follow either a static or dynamic method. In the static method, stations are prioritized based on calculated focal points of the operational areas, considering activation time depending on station type. In the dynamic method, the simulation selects the vehicle that can reach the incident location fastest in terms of travel time. Once selected, the EVs are activated and proceed through their respective operational workflows. For professional fire departments, personnel first put on their gear before departure, then vehicles follow a realistic route to the incident, which is handled by the SUMO traffic simulation. After an on-scene operation, they become available for the next incident and return to their station. Volunteer fire departments follow a similar procedure, but their personnel must first travel from distributed home locations to the station before gearing up, and return home conceptually after the mission. Ambulances also include preparation time before departure, provide on-scene treatment, and then transport the patient to a hospital, typically the nearest available one, unless capacity or special care requirements necessitate a different destination. After delivering the patient, the ambulance returns to its base and becomes available for further incidents.

4. Case Study: Munich Fire Department

The presented simulation and analysis platform from Section 3 is now applied to a case study with the Munich Fire Department. The following software versions were used for this research: PostGIS 3.5.2, pgRouting 3.7.3, and SUMO 1.21.0.

Data recording: Fire Station Four of the Munich Fire Department was recorded from February 2024 to May 2024 and Fire Station One was recorded from June 2024 to May 2025. At both fire stations the standard formation of the Munich Fire Department consisting of a command vehicle (CV), two fire engines (FE), a turntable ladder (TL) and an ambulance (A) are included in the recording. Additionally, the vehicles of the operations control service and an additional ambulance at Fire Station One were also recorded. The EVs can be divided into a small vehicle class (CV and A) and a large vehicle class (FE and TL). For recording purposes, the vehicles were equipped with a data logger connected to the vehicle's OBD port, which recorded both GPS information and the IMU signal. Each data logger consists of an ESP-32 microcontroller, a MAX-M8W GNSS module from ublox, and an external GNSS antenna [27]. The extent of the recorded data can be seen in Figure 4.

Road network: The road network was exported from [30]. The data was afterwards converted to a car-only SUMO network using netconvert of [8]. Frequently used street segments not included within the initial export were added manually. The road network was further enriched by labeling the streets' mobility-transition-related attributes that may influence EVs. The type of cycling infrastructure and the presence of dedicated special lanes, such as tram tracks that can be driven on, were manually classified into the following categories based on aerial photographs and road surveys:

- **Cycling infrastructure:** Presence of cycle lanes accessible to cars without a curb
- **Separation of traffic directions:** no structural separation (only markings); structural separation (guard rails, central reservation);
- **Special lanes:** special lanes are available and accessible to EVs (e.g., bus lanes).

Traffic data: Traffic data is obtained on a minute-by-minute basis from [28,29] and archived in the database. The traffic data is referenced using a geometric assignment in which equidistant points are generated at intervals of 10 m along the internal network edges. For each of these points, the nearest external network link within a comparably small threshold is determined, accounting for both distance and travel direction. An assignment is only valid if the point lies within the link and the distance as well as directional deviation are below a defined threshold value.

The quality of the matching was ensured and checked using the user interface.

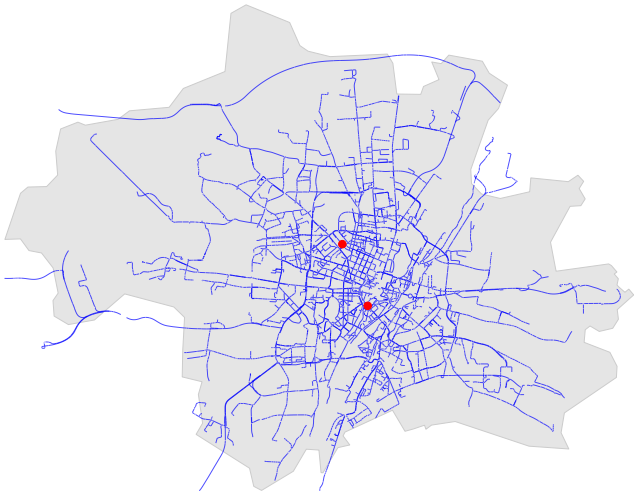


Figure 4. Edges with recorded data are shown in blue. Fire Stations One and Four are marked as red circles (updated version of own figure from [27]).

5. Results

This section presents the results of the case study. The section is divided into the results of the data analysis and the simulation case study.

5.1. Results of the Data Analysis

The data collected in our case study (see Section 4) was processed using the approach described in Section 3 to calculate the previously defined indicators $R_{\text{Roadsegment}}$ and perform a regression analysis to determine the relationship between the speed of EVs and the average traffic speed on each road segment.

In the following subsections, first the indicators are discussed to demonstrate the plausibility and their feasibility for identifying problematic edges. Second, the influence of the structural separation between the driving directions is discussed. Finally, five different road cross-sections are discussed in detail.

5.1.1. General Indicator Analysis

First, we descriptively examine the regressions introduced in Section 3.1.2 for all street segments. Figure 5 summarizes the resulting distributions of slope, intercept and EV speed advantage for large vehicles (LV, FE/TL class) in Figure 5a–c and small vehicles (SV, CV/A class) in Figure 5d–f. For both classes, the slope distribution is predominantly positive and below 1. The LV slope distribution is visibly shifted to higher values than the SV distribution. The intercept distributions—formally the model's EV speed at $v_{\text{Traffic}} = 0$ should be interpreted with caution because observations with low observed traffic speeds are rare. They serve primarily to position the regression line. Finally, the EV speed advantage $v_{\text{EV,adv}} = \frac{v_{\text{EV}}}{v_{\text{Traffic}}}$ is almost always above 1 but widely dispersed for both classes, reflecting heterogeneous local conditions. The advantage is larger and more stable for SV than for LV. The plausibility of the compound criticality indicator was reviewed with domain experts, who confirmed that locations flagged align with known operational bottlenecks. The segment highlighted in Figure 6 exhibits, for example, a high criticality. As shown by the street-level image in (b), construction work reduces the carriageway to a barrier-bounded one-lane corridor.

5.1.2. Discussion of the Influence of Structural Separations Between Traffic Directions

This section investigates whether the physical separation of opposite driving directions increases the likelihood of low-speed occurrences ($v < 2\,\frac{\text{m}}{\text{s}}$) on two-lane roads. To ensure data quality and comparability, the first and last 20 s of each trajectory were truncated to avoid start and stop effects for this analysis. Only two-lane road segments with a minimum length of 10 m were considered, and segments with drivable bicycle lanes were excluded. Furthermore, a minimum of 10 valid measurements per segment was required for inclusion in the analysis. The probability of critical events was compared between road segments with and without physical separation of driving directions visible in Figure 7. Performing a Mann–Whitney U test reveals a statistically significant difference between the two groups for both vehicle classes. These results indicate that road segments with physically separated driving directions exhibit a lower probability of low-speed events than those without separation. Additionally, the regression analysis presented in Section 3.1.2 was conducted to investigate whether the speed of EVs and the reported average speed of surrounding traffic vary depending on the type of separation. For both conditions, Theil-Sen regression models were fitted, and slopes and intercepts were compared. The strength of the linear relationship was quantified using Pearson's r. The distribution of these regression parameters for large vehicles and for small vehicles is shown in Figure 8. Similar patterns are evident in the slope and intercept distributions for both vehicle classes. Where oncoming traffic is physically separated, the slopes are higher and the intercepts lower. Taken together, our data shows that on edges with direction separation the association of the speed of EVs and surrounding traffic is stronger. Additionally, the relationship is stronger for the large vehicle class.

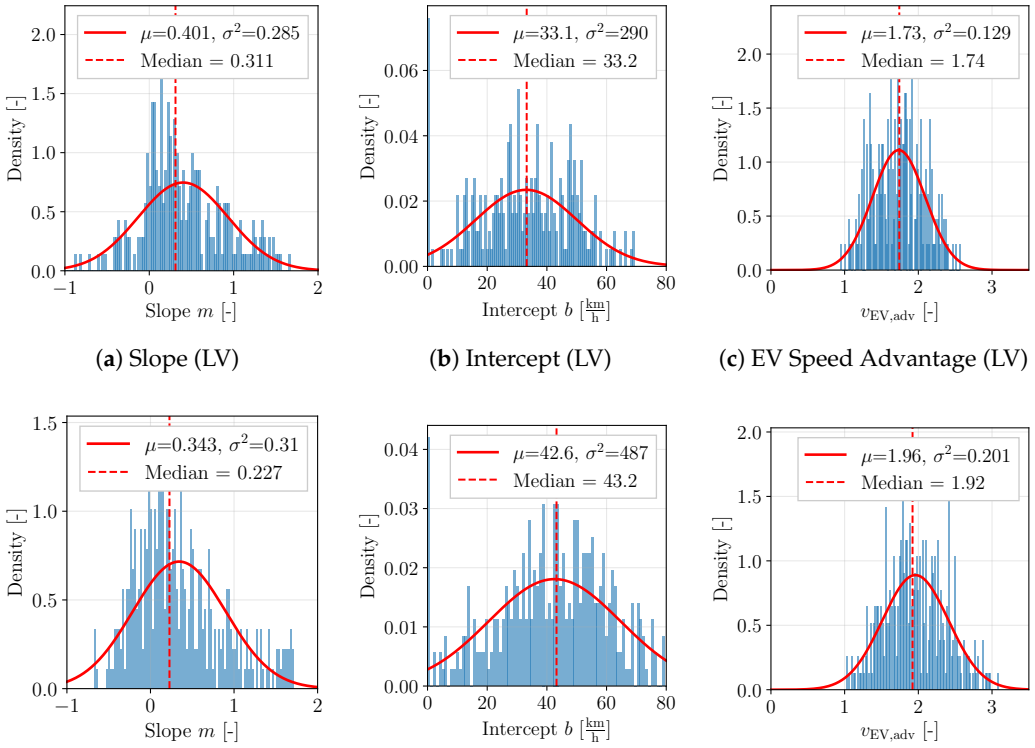
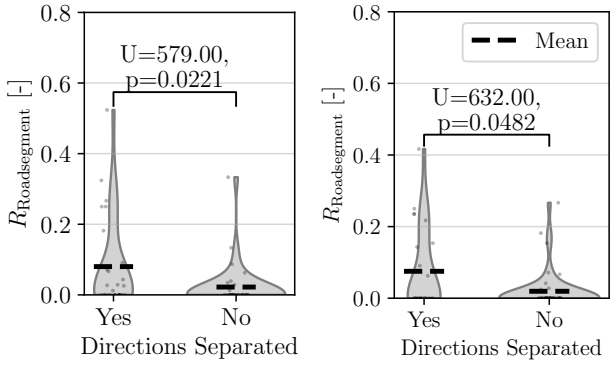


Figure 5. Distributions for regression analysis indicators. (**Legend:** LV = large vehicles; SV = small vehicles).



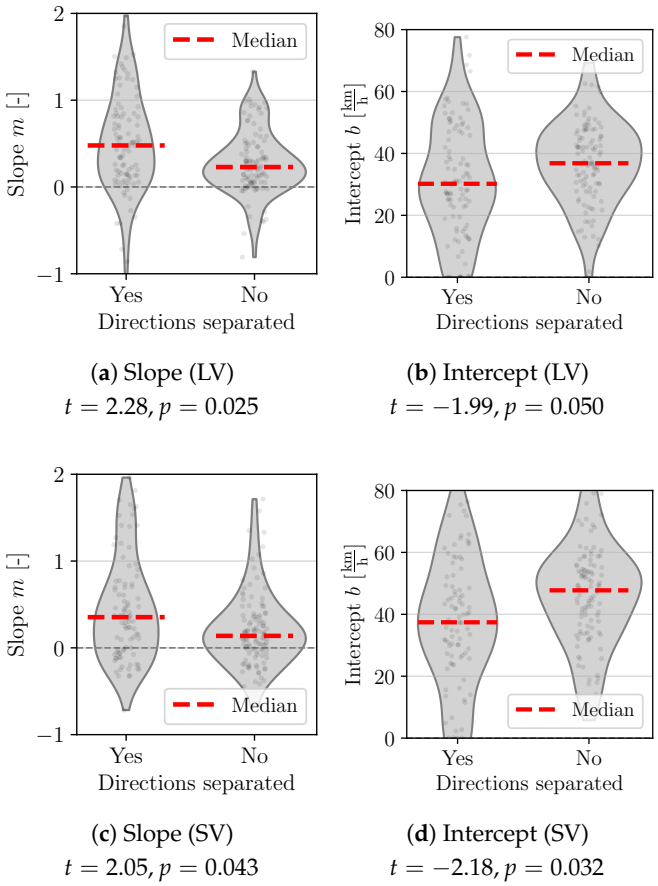
(**a**) (**b**)

Figure 6. Most critical recorded link in (**a**) Network (Source: [8]) and (**b**) Street-level Image (Source: authors) representation.



(**a**) Risk of getting stuck (LV) (**b**) Risk of getting stuck (SV)

Figure 7. Relative Criticality Index $R_{\text{Roadsegment}}$ focusing on low-speed occurrences ($v < 2\,\frac{\text{m}}{\text{s}}$) by road segment attributes (**Legend:** LV = large vehicles; SV = small vehicles).



(**a**) Slope (LV)
$t = 2.28, p = 0.025$

(**b**) Intercept (LV)
$t = -1.99, p = 0.050$

(**c**) Slope (SV)
$t = 2.05, p = 0.043$

(**d**) Intercept (SV)
$t = -2.18, p = 0.032$

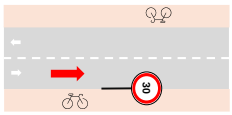
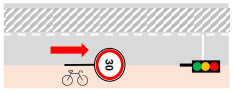
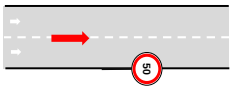
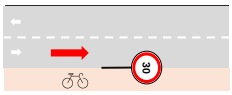
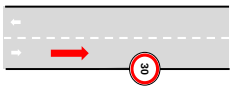
Figure 8. Regression analysis for small and large EVs. (**Legend:** LV = large vehicles; SV = small vehicles).

5.1.3. Discussion of Various Road Cross-Sections

Finally, five different road sections with many records will be examined in detail. The results are shown in Table 2. For each section, we perform a regression of the EV speed against the nearest available reported average traffic speed and indicate the slope, the intercept, the sample size N, and the observed range of reported traffic speeds. Across all sections, N is large enough to provide stable Theil-Sen estimates, and the width of the traffic speed range indicates the range of traffic conditions under which the data for this link was collected. Across the five representative cross-sections, a consistent pattern emerges. Where evasion options are structurally limited, i.e., where there are no passable

cycle lanes and no usable opposite lanes, as in Sections 3 and 5, the coupling between the speed of EVs and background traffic is stronger, which is reflected in higher slopes and larger Pearson correlation coefficients r. This limitation is even more pronounced for the LV class. In contrast, cross-sections that offer usable evasion space, such as Section 1 with a passable opposite lane and passable cycle lanes, show very low gradients for both classes.

Table 2. Comparison of slope, intercept, number of measurements (N), FCD width and bootstrap analysis between small and large vehicles for each road cross-section. Bootstrap column shows median slope, 95% CI, and bootstrap p-value. (**Legend:** * = $p < 0.05$).

Road Section	Small Vehicles						Large Vehicles					
	m	b	N	Range $v_{Traffic}$	Pearson r	Bootstrap $\tilde{m}$, CI, p	m	b	N	Range $v_{Traffic}$	Pearson r	Bootstrap $\tilde{m}$, CI, p
	0.10	45.98	208	11–34	−0.036	0.100, [−0.115, 0.305], 0.344	−0.07	47.42	254	12–34	−0.092	−0.043, [−0.283, 0.227], 0.752
	0.28	46.93	212	10–34	0.113	0.320, [0.087, 0.566], 0.008	0.07	49.58	252	11–34	−0.025	0.085, [−0.152, 0.335], 0.436
	0.78	34.57	91	11–42	0.247 *	0.797, [0.168, 1.378], 0.008	0.82	28.20	121	10–45	0.357 *	0.856, [0.528, 1.341], 0.000
	0.19	43.25	65	13–35	−0.059	0.239, [−0.310, 0.804], 0.360	0.06	44.18	101	12–35	−0.071	0.085, [−0.187, 0.317], 0.496
	0.80	26.51	95	14–34	0.243 *	0.823, [0.388, 1.450], 0.000	0.99	20.81	141	9–32	0.338 *	1.005, [0.638, 1.502], 0.000

The results of the data analysis can be summarised as follows: More favorable traffic conditions are associated with improved progress of EVs. In addition, road geometry also plays a role: undivided roads are associated with a lower likelihood of low-speed events. Our analysis further indicates that physically separated corridors exhibit a stronger association between surrounding traffic speed and EV speed. These findings are observational and should not be interpreted as causal effects.

5.2. Results of the Dynamic System Model

In the following section of this paper, the case study with the developed agent-based model is presented. In the first step, the model is evaluated on its overall quality, with a particular focus on the temporal accuracy of EV trips and dispatch performance. This is followed by a sensitivity analysis to show the effects of the network parameterization on the model. The underlying system setup is applied as shown in Figure 9 and described in Section 4. The model includes the ten Munich Fire Stations and all public hospitals (based on the OSM tags *amenity* = *hospital* and *emergency* = *yes*) in Munich exported from [30]. The reference control center data set is March 2024. The alarm and dispatch order was derived based on the most frequently dispatched vehicle configuration for each incident keyword.

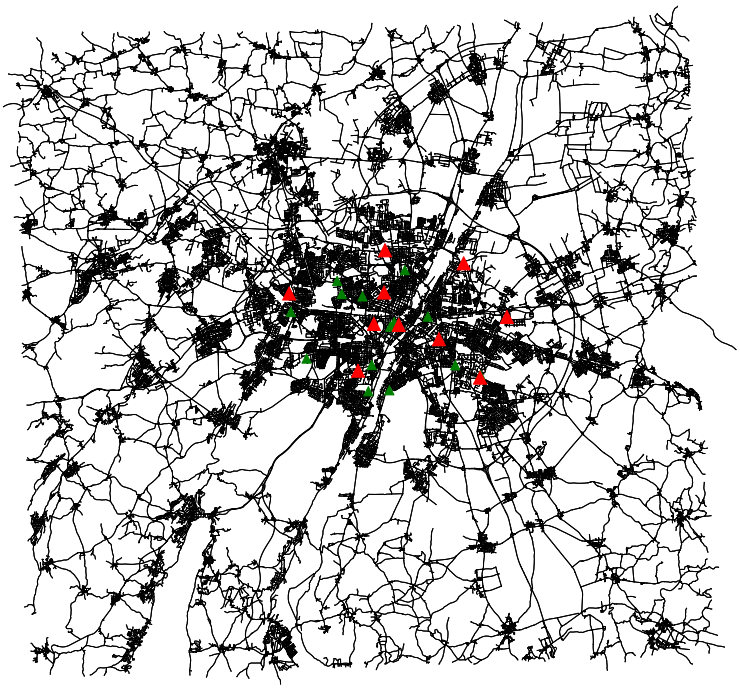


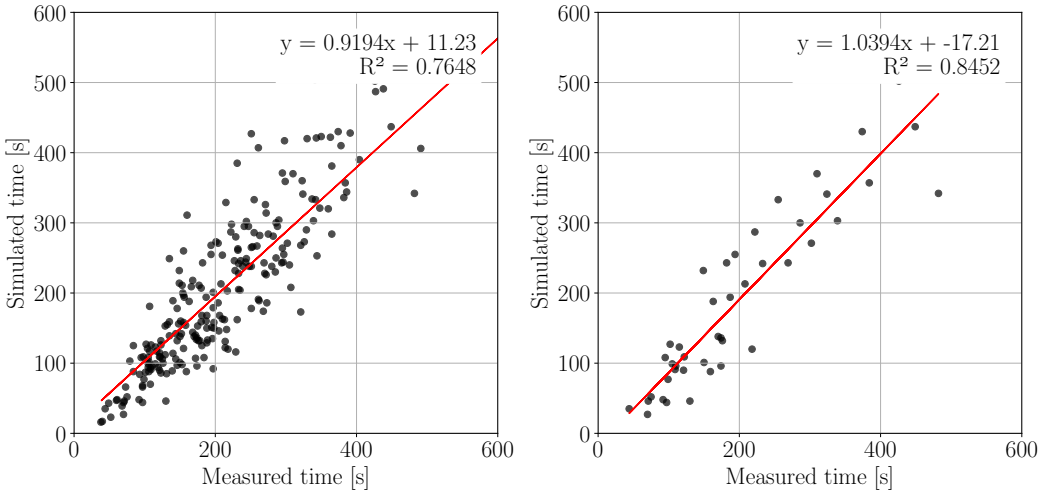
Figure 9. Setup for system validation (**Legend:** red triangle = Fire Station, green triangle = Hospital).

5.2.1. Validation

In a first step the dynamic system model is validated against observed travel times. To evaluate the temporal accuracy of the model, the analysis focuses on the first-arriving large EV. Historical data from the control center protocol is frequently used for strategic planning. Accordingly, the validation step compares the simulated travel times with the recorded travel times (status 3 to status 4) derived from the control center protocol. Prior to validation, the data set was checked and filtered to ensure its plausibility and relevance for the validation data set, e.g., by removing empty or incorrect entries that may occur due to radio-based reporting. The validation results for time-critical incidents are shown in Figure 10. The regression line in Figure 10a exhibits a slope of 0.9194 and an intercept of 11.23 s. It shows that the model slightly underestimates the travel times of the first-arriving EVs for longer trips. The mean absolute percentage error (MAPE) of the overall simulation is 21.91%, while the mean absolute error (MAE) is 42.61 s. The error analysis grouped by real world trip duration shows that short trips ($t < 180$ s) have a MAPE of 25.19% and a MAE of 30.49s. Medium trips ($180 \leq t < 420$ s) exhibit a MAPE of 19.50% and an MAE of 49.12 s , while long trips ($t \geq 420$ s) have a MAPE of 17.17% and an MAE of 102.29 s . In 85.3% of all cases, the simulation correctly predicts the first-arriving large EV, as a result of dispatching and simulation. In 14.7%, the first-arriving large EV differs from reality.

The observed deviations in travel time can be attributed to two causes, among others: (i) missing roads in the underlying road network, which is cars-only, leading to unrealistic routes in areas with pedestrian zones and access restrictions, and (ii) local incorrect estimates of travel speeds and junction time losses due to the use of average values for parameterization. Especially, (i) leads to spatially clustered delays in travel time.

While we have recorded GPS data for network calibration for Fire Stations One and Four, it is not available for all other fire stations. Figure 10b shows the validation result for Fire Station Seven using the average historical speeds. This shows (to a certain extent) that average speeds measured at one station can be transferred to another and used to produce a suitable model.



(**a**) All stations (**b**) Fire Station Seven

Figure 10. Validation results of the dynamic system model.

5.2.2. Sensitivity Analysis

The following section focuses on examining the sensitivity of the system response to locally altered speeds. Motivated by the conducted data analysis, which shows the strategic role of main urban roads for EVs, we generate three interruption scenarios by randomly selecting 20% of all main-road segments in Munich and reducing the EV travel speed on those links by 50%. Selections are independent across scenarios and may affect links on the way to the incident or to the hospital. No modal filters or closures are introduced. The outcomes in Table 3 show a system-wide sensitivity. Across all interruption scenarios, both the medians and the mean values increase for response time of large EVs to incidents and the total travel time of ambulances to the hospital.

In short, even modest, spatially scattered speed reductions on main roads propagate into system-wide response-time increases. These findings underscore the need to monitor and analyze planned speed reductions, narrowings, or construction phases specifically against established EV corridors, for example by using the indicators in Section 3.1.2.

Table 3. Travel and Response Times: Incident and Hospital.

Scenario	Response Time of Large EVs to Incidents [s]		Total Travel Time to Hospital [s]	
	Median	**Mean**	**Median**	**Mean**
Base Scenario	244	297	576	662
Interruption 1	277	337	658	749
Interruption 2	270	330	660	744
Interruption 3	271	332	647	739

6. Discussion

The framework was developed to support emergency services during the transition to sustainable mobility to handle the challenges that come with it. Although the results show clear benefits of our method in monitoring the current emergency response system and planning future scenarios, there are still limitations and priorities for future work.

The following section deals with the most important limitations:

- *Firstly*, the underlying data set is limited. Our analysis is based on a limited observation window from two out of ten stations of the Munich Fire Department. Although this data set is comprehensive and detailed compared to the literature, it nevertheless

represents only a snapshot of a single city. The transferability of the results to other urban contexts needs to be verified by applying the framework to other fire stations and cities.

- *Secondly*, the regression analysis is sensitive to the quality of the input data. Both network-to-network referencing and the data, e.g., the FCD-reported average speed used as an indicator for the traffic condition, itself introduce errors. Future data collection should supplement the logger data with actual scene recordings, for example, through ego-perspective, geo-referenced camera recordings. This would enable a precise reconstruction of the traffic situation including traffic light states and the observed behavior of the surrounding road users encountered by the EV. This would allow driving behavior and conflict situations to be modeled more accurately, and the causes for observed values to be investigated. The regression-based speed-coupling indicator was designed as a simple, edge-wise and univariate measure. Therefore, multicollinearity is not of concern here. Heteroskedasticity is handled by reporting bootstrapped uncertainty for the Theil–Sen slope, including confidence intervals. Some spatial dependence between adjacent links is expected in road networks, however, the indicator is computed separately for each topologically defined link. We therefore treat link-to-link tests as descriptive.
- *Thirdly*, the current assessment is based on geometric simplifications. To reduce distortions, future work should take into account lane width, geometry at lane level, and the exact intersection layouts. Therefore, the detailed relationship between structural design decisions and operational safety, including the perspective of vulnerable road users, remains an open question.
- *Fourthly*, the simulation model offers an accessible, end-to-end approach for realistically simulating ground-based rescue operations at the system level; however, it is not yet complete. Air medical services have not been integrated, which limits coverage of the emergency response system to ground-based vehicles. Processes and interactions at the scene, such as extrication in road-traffic collisions, are currently out of scope, even though they can materially influence both duration and outcomes. The target variable in the present validation is arrival at the presumed incident location; in practice, arrival at the actual patient location is more relevant and should be reflected in future versions. A comprehensive validation and application in the context of a case study for volunteer fire departments remains outstanding and should be completed to ensure robustness across organizational types.
- *Fifthly*, the simulation model uses a mesoscopic traffic model that does not explicitly simulate surrounding traffic. Therefore, it does not consider congestion caused by bottlenecks or the redistribution of traffic within the network. Nevertheless, detailed simulations under changed traffic conditions require a microscopic simulation of surrounding traffic to capture diversion and congestion effects. While this can be achieved with tools such as *hybridPY* and the microscopic fire brigade model [16,43], it presupposes an underlying traffic demand model. Incorporating such an assignment where warranted will improve fidelity for scenario analyses that include widespread speed changes, construction phases, or modal filters.
- *Finally*, the model has been tailored to the Bavarian context; its applicability to other countries has not yet been tested and will likely require adapting institutional rules, dispatch logic, and infrastructure constraints. In order to transfer the model to other cities, the underlying database can be derived from publicly available data and specific data from emergency service operators, e.g., station locations and incident data, can be integrated to perform initial analyses. Detailed data, e.g., GPS measurements for traffic

analysis, can then be added in a second step. Thus, the model centres on globally transferable components enriched with city-specific data.

In summary, the framework already enables data-driven analysis and planning. However, more comprehensive data sets, richer environmental sensing, more detailed geometry, advanced process modelling, and assignable traffic dynamics are the levers to strengthen validity, portability, and decision value.

7. Conclusions

This paper presents a holistic, open-source-based, and modular simulation and analysis platform that enables emergency services to monitor the interaction between the transport system and the emergency response system and efficiently investigate 'what-if' scenarios. The platform links heterogeneous data sources—real-time traffic data, GPS recordings from EVs and an OSM-based road network [30]—in a data model and uses pgRouting [32] for indicator-based network analyses and SUMO [8] for mesoscopic, agent-based simulations. The agent-based system replicates all relevant ground-based actors in the emergency response systems. The presented framework helps organizations to identify road sections that require manual inspection and investigate changes at the system level, whether due to changes to the transport system or organizational measures.

The case study with the Munich Fire Department demonstrates the practical applicability of the framework. Within this case study, critical road sections are identified using the combined criticality indicator. In a second step, the influence of road geometry and physical separation on the progress of EVs in urban environments is discussed. More favorable traffic conditions are associated with improved progress of EVs and undivided roads are associated with a lower likelihood of low-speed events. Our data further indicates that physically separated corridors exhibit a stronger association between the surrounding traffic speed and the EV speed.The developed agent-based model shows promising validation results when compared to historical incident data.

Nevertheless, several limitations remain. Air medical services are not yet included, processes at the scene are simplified, and the transferability to other regions needs to be verified. Future work should extend the underlying data set, add additional data such as ego-perspective video, and incorporate richer geometric detail where needed. Overall, the framework provides a solid basis for data-driven decision-making in emergency response system planning.

Author Contributions: Conceptualization, F.S.; methodology, F.S.; software, F.S., M.S. and T.Z.; validation, F.S. and T.Z.; formal analysis, F.S.; investigation, F.S.; resources, M.L.; data curation, F.S. and T.Z.; writing—original draft preparation, F.S., M.S. and T.Z.; writing—review and editing, F.S., M.S. and T.Z.; visualization, F.S. and T.Z.; supervision, M.L.; project administration, F.S.; funding acquisition, M.L. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The underlying SUMO networks originate from freely accessible and usable OpenStreetMap data extracts. The historical traffic data used in this study were obtained from the City of Munich and are currently accessible via mobilithek.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

A	Ambulance
CV	Command vehicle
EV	Emergency vehicle
eVTOL	Electric vertical take-off and landing aircraft
FCD	Floating car data
FD	Fire Department
FE	Fire engine
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
IMU	Inertial measurement unit
LV	Large vehicles
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
OBD	On-board diagnostics
OSM	OpenStreetMap
pgRouting	PostGIS routing extension
PostGIS	PostgreSQL geospatial extension
SUMO	Simulation of Urban MObility
SV	Small vehicles
TL	Turntable ladder vehicle
VFD	Volunteer Fire Department

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